

JEE ADVANCED MOCK TEST - 1

PAPER 2: MATHEMATICS

SECTION - I (Multi-Select Choice Type)

This section contains 8 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE** choice is correct.

Marking Scheme:

- **Full Marks (+4):** If only the correct options are chosen.
- **Partial Marks (+3):** If all four options are correct, but only three are chosen.
- **Partial Marks (+2):** If three or more options are correct, but only two correct ones are chosen.
- **Partial Marks (+1):** If two or more options are correct, but only one correct option is chosen.
- **Zero Marks (0):** If the question is left unattempted.
- **Negative Marks (-2):** If any incorrect option is selected.

1. Let the function $f(x) = \sum_{n=1}^3 ((\sin^2 x)^n + (\cos^2 x)^n)$ for all $x \in \mathbb{R}$, then:
 - (A) The maximum value of $f(x)$ is $\frac{5}{2}$
 - (B) The maximum value of $f(x)$ is 3
 - (C) The minimum value of $f(x)$ is $\frac{7}{4}$
 - (D) The minimum value of $f(x)$ is $\frac{9}{4}$
2. The number of solutions of the equation $\sin^3 x \cos x + \sin^2 x \cos^2 x + \sin x \cos^3 x = 1$ in the interval $[0, 2\pi]$ is not equal to:
 - (A) 4
 - (B) 2
 - (C) 1
 - (D) 0
3. For $a > 1$, $b > 1$, it is given that $7(\log_b a - \log_a b) = 48$ and $a = 8b$. If $a - 2b = 3\sqrt{k}$, then k is a/an:
 - (A) Even natural number
 - (B) Odd natural number
 - (C) Perfect cube of a natural number
 - (D) Perfect square of a natural number
4. For the functions $f(x) = (x^2 + bx + c)e^x$ and $g(x) = (x^2 + bx + c)e^x + e^x(2x + b)$, which of the following holds good?
 - (A) If $f(x) > 0$ for all real $x \implies g(x) > 0$ for all real x
 - (B) If $f(x) > 0$ for all real $x \implies g(x) > 0$ for all real x
 - (C) If $g(x) > 0$ for all real $x \implies f(x) > 0$ for all real x
 - (D) $g(x) > 0$ for all real $x \iff f(x) > 0$ for all real x

5. The possible value(s) of n such that

$$\frac{2}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^\circ \sin 48^\circ} + \frac{1}{\sin 49^\circ \sin 50^\circ} + \cdots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \frac{1}{\sin n^\circ}$$

is/are:

- (A) 179
 - (B) 181
 - (C) 359
 - (D) 361
6. The value(s) of x satisfying the equation $\frac{1-2(\log_{10} x)^2}{\log_{10} x - 2(\log_{10} x)^2} = 1$ is/are:
- (A) $\frac{1}{\sqrt{10}}$
 - (B) $\sqrt[3]{10}$
 - (C) $\frac{1}{\sqrt{20}}$
 - (D) $\sqrt{10}$
7. If $x + y + z = xyz$, then the expression $\frac{x}{\sqrt{1+x^2}} + \frac{y}{\sqrt{1+y^2}} + \frac{z}{\sqrt{1+z^2}}$ cannot be equal to:
- (A) 2
 - (B) 3
 - (C) 4
 - (D) 5
8. For all $\lambda \in \mathbb{R}$, the equation $ax^2 + (b - \lambda)x + (a - b - \lambda) = 0$ ($a \neq 0$) has real roots. Then which of the following may be true?
- (A) $a = b$
 - (B) $b < a < 0$
 - (C) $b > a > 0$
 - (D) $a > b > 0$

SECTION - II (Paragraph / Comprehension Type)

This section contains **2 Paragraphs**. Each paragraph has **2 Multiple Choice Questions** based on it. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** choice is correct.
Marking Scheme: +4 for a correct answer, -1 for an incorrect answer, 0 if unattempted.

Paragraph for Question 9 and 10

Consider the following system containing real parameters α, β, γ :

$$\log_2(\alpha^6 - 16\alpha^3 + 66) + \sqrt{4\beta^4 - 8\beta^2 + 13} + \left| \frac{\gamma}{3} - 2 \right| = 4$$

9. The number of unique ordered triplets (α, β, γ) satisfying the given relation is:

- (A) 0
- (B) 2
- (C) 4
- (D) 6

10. The sum of all possible values of the product $\alpha\beta\gamma$ is equal to:

- (A) 0
- (B) 6
- (C) 12
- (D) 24

Paragraph for Question 11 and 12

Consider the logarithmic equality given below:

$$\log_{10} |x^3 + y^3| - \log_{10} |x^2 - xy + y^2| + \log_{10} |x^3 - y^3| - \log_{10} |x^2 + xy + y^2| = \log_{10}(221)$$

11. If $y = 2$, then the value of x can be:

- (A) ± 111
- (B) ± 15
- (C) ± 2
- (D) ± 110

12. If $x + y = 17$, then the sum of all possible values of x is:

- (A) 16
- (B) 15
- (C) 14
- (D) 17

SECTION - III (Numerical Value / Integer Type)

This section contains **6 Numerical Value Type Questions**. The answer to each question is an integer.

Marking Scheme: +4 for a correct answer, -1 for an incorrect answer, 0 if unattempted.

13. The value of $\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7}$ is equal to _____.
14. The smallest value of $(8 \cos^2 k)$ for which the equation $\sin x + \cos(k+x) + \cos(k-x) = 2$ has real roots is _____.
15. The smallest integral value of x satisfying the equation:

$$\log_4(6 + \sqrt{x} - |\sqrt{x} - 2|) = \frac{1}{2} + \log_2 |\sqrt{x} - |\sqrt{x} - 2||$$
 is _____.
16. The number of negative integral values of x satisfying the inequality $\log_{0.6} \left(\log_6 \frac{x^2+x}{x+4} \right) < 0$ is _____.
17. For positive numbers a, b, c , the minimum value of $\frac{a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2)}{abc}$ is _____.
18. If $n \in \mathbb{I}$ and the roots of the quadratic equation $x^2 + 2nx - 19n + 92 = 0$ are rational in nature, then the sum of all possible distinct values of n is equal to _____.